📄 **Mini Project Documentation**

🧩 **Project Title: Task Management / Board Application**

🔗 **Live URL**

Open Project

---

## 📌 1. Introduction

This project is a **Task Management Web Application** where users can organize their work using boards, lists, and cards (similar to Trello). It helps users manage tasks efficiently, track progress, and collaborate.

---

## 🎯 2. Objectives

- To provide a platform for managing tasks visually
- To organize work into boards and lists
- To allow users to create, update, and delete tasks
- To improve productivity and workflow management

---

## 🛠️ 3. Tech Stack

- **Frontend:** HTML, CSS, JavaScript (possibly React)
- **Backend:** Node.js, Express
- **Database:** MongoDB
- **Deployment:** Vercel

➡️ Projects on platforms like Vercel are deployed apps connected to a Git repository and can have multiple deployments and configurations.

---

## ⚙️ 4. Features

### 👤 User Features

- User authentication (Login / Signup)
- Session-based access control

📋 **Board Management**

- Create multiple boards
- Each board represents a project

🗂️ **List Management**

- Create lists inside boards (e.g., To Do, In Progress, Done)

📝 **Card/Task Management**

- Add tasks inside lists
- Edit task details
- Delete tasks

🔄 **Drag & Drop (if implemented)**

- Move tasks between lists

---

📐 **5. System Architecture**

Client (Browser)

↓

Frontend (UI)

↓

Backend API (Express)

↓

Database (MongoDB)

---

🔐 **6. Authentication Flow**

1. User logs in
2. Server verifies credentials
3. Session is created
4. Protected routes are accessed using middleware

Example:

```
function requireAuth(req, res, next) {
  if (!req.session.userId) {
    return res.redirect('/auth/login');
  }
  return next();
}
```

---

### 🧪 7. API Endpoints (Sample)

**Auth**

- POST /auth/register
- POST /auth/login

**Boards**

- GET /boards
- POST /boards
- DELETE /boards/:id

**Lists**

- POST /lists
- DELETE /lists/:id

**Cards**

- POST /cards
- PUT /cards/:id
- DELETE /cards/:id

---

### 💾 8. Database Schema (Basic)

**User**

- id
- email
- password

**Board**

- id
- title
- userId

**List**

- id
- title
- boardId

**Card**

- id
- title
- description
- listId

---

### 🎨 9. UI Overview

- Dashboard showing all boards
- Board view with multiple lists
- Cards inside lists
- Clean and interactive interface

---

### 🚀 10. Deployment

- Hosted on **Vercel**
- Continuous deployment using GitHub
- Accessible via public URL

---

### 📈 11. Advantages

- Easy task tracking
- Visual workflow management

- Improves productivity
- Scalable architecture

---

### ⚠️ 12. Limitations

- May require login to access boards
- Limited collaboration features (if not implemented)
- No real-time updates (if sockets not used)

---

### 🔮 13. Future Enhancements

- Real-time collaboration (Socket.io)
- Notifications system
- File attachments
- Team collaboration features
- Mobile responsiveness improvements

---

### 🏁 14. Conclusion

This project demonstrates full-stack development skills including frontend UI design, backend API creation, database management, and deployment. It is a practical implementation of a real-world productivity tool.